

1. $\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$
 $\chi^2_{1-\alpha/2} < \frac{(n-1)s^2}{\sigma^2} < \chi^2_{\alpha/2}$ in order to make it an increasing function of σ^2 , take the inverse of all sides

$$\frac{1}{\chi^2_{1-\alpha/2}} < \frac{\sigma^2}{(n-1)s^2} < \frac{1}{\chi^2_{\alpha/2}} \text{ now multiply everything by } (n-1)s^2$$

$$\frac{(n-1)s^2}{\chi^2_{1-\alpha/2}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{\alpha/2}}$$

2. $\frac{2Y}{\beta} \sim \chi^2_4$
 $\chi^2_{4,0.95} < \frac{2Y}{\beta} < \chi^2_{4,0.05}$ in order to make it an increasing function of β , take the inverse of all sides

$$\frac{1}{\chi^2_{4,0.95}} < \frac{\beta}{2Y} < \frac{1}{\chi^2_{4,0.05}} \text{ now multiply everything by } 2Y$$

$$\frac{2Y}{\chi^2_{4,0.95}} < \beta < \frac{2Y}{\chi^2_{4,0.05}}$$

The R code below calculates the chi-squared quantiles at 0.95 and 0.05 with 4 degrees of freedom

```
qchisq(0.95,4)
```

```
## [1] 9.487729
```

```
qchisq(0.05,4)
```

```
## [1] 0.710723
```

$$\frac{2Y}{9.487729} < \beta < \frac{2Y}{0.710723} \rightarrow \left(\frac{2Y}{9.487729}, \frac{2Y}{0.710723} \right)$$

3. b) $2 \sum_{i=1}^n \frac{Y_i}{\theta} \sim \chi^2_{2n}$
 $\chi^2_{2n,0.975} < 2 \sum_{i=1}^n \frac{Y_i}{\theta} < \chi^2_{2n,0.025}$
 $\chi^2_{2n,0.975} < \frac{2\bar{Y}n}{\theta} < \chi^2_{2n,0.025}$ in order to make it an increasing function of θ , take the inverse of all sides

$$\frac{1}{\chi^2_{2n,0.975}} < \frac{\theta}{2\bar{Y}n} < \frac{1}{\chi^2_{2n,0.025}} \text{ now multiply everything by } 2\bar{Y}n$$

$$\frac{2\bar{Y}n}{\chi^2_{2n,0.975}} < \theta < \frac{2\bar{Y}n}{\chi^2_{2n,0.025}} \rightarrow \left(\frac{2\bar{Y}n}{\chi^2_{2n,0.975}}, \frac{2\bar{Y}n}{\chi^2_{2n,0.025}} \right)$$

c) The R code below calculates the chi-squared quantiles at 0.975 and 0.025 with 14 degrees of freedom.

```
qchisq(0.975,14)
```

```
## [1] 26.11895
```

```
qchisq(0.05,14)
```

```
## [1] 5.628726
```

Then plugging in $n=7$, $\bar{y} = 4.77$, and $\chi^2_{2n,0.975} = 26.11895$, $\chi^2_{2n,0.025} = 5.628726$ results in $(\frac{2(4.77)(7)}{26.11895}, \frac{2(4.77)(7)}{5.628726}) = (2.5568, 11.8641)$

$$4. \text{ a) } \hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

The R code below calculates the z critical value at 0.01.

```
qnorm(0.01,lower.tail=FALSE)
```

```
## [1] 2.326348
```

$$0.45 \pm 2.326248 \sqrt{\frac{0.45(1-0.45)}{800}}$$

$$0.45 \pm 2.326248(0.0175891)$$

$$0.45 \pm 0.0409 \rightarrow (0.4091, 0.4909)$$

We are 98% confident that the true proportion of adults that think movies are getting better is between (0.4091, 0.4909).

b) The interval does not include $p=0.50$. Then the majority of adults do not say that movies are getting better.

$$5. \bar{x}_1 - \bar{x}_2 \pm t_{\alpha/2, n_1+n_2-2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

The R code below calculates $t_{\alpha/2, n_1+n_2-2}$ for $\alpha=0.05$ and $n_1 + n_2 - 2 = 34$

```
qt(0.025,34, lower.tail=FALSE)
```

```
## [1] 2.032245
```

$$11-12 \pm 2.032245 \sqrt{\frac{6^2}{16} + \frac{8^2}{20}}$$

$$-1 \pm 2.032245(2.334524)$$

$$-1 \pm 4.7443$$

$$(-5.7443, 3.7443)$$

We are 95% confident that the true difference in the first drug and second drug's effects on lowering blood pressure is between $(-5.7443, 3.7443)$ points. Since 0 is in the interval, we cannot conclude that there is a significant difference between the two drugs.